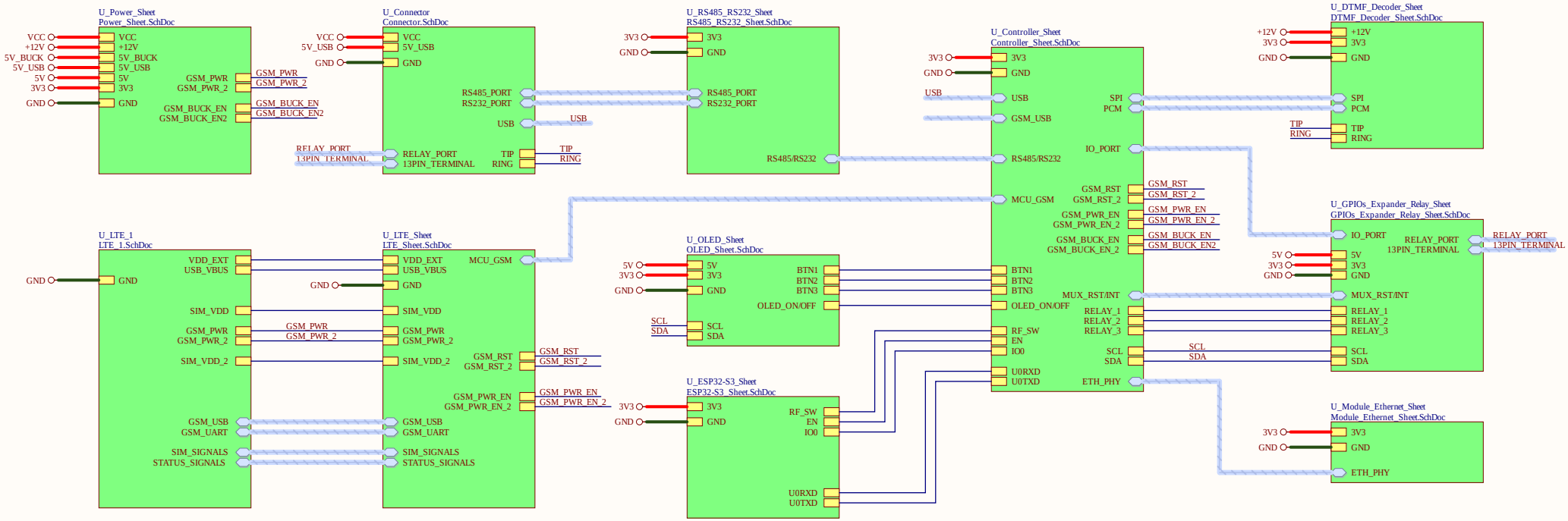


Option 1: (Single Path - Ethernet capability only)
---> DNP U13, U16, U17, J21, J23.

Option 2: (Single Path - LTE capability only)
---> DNP J20, U12.

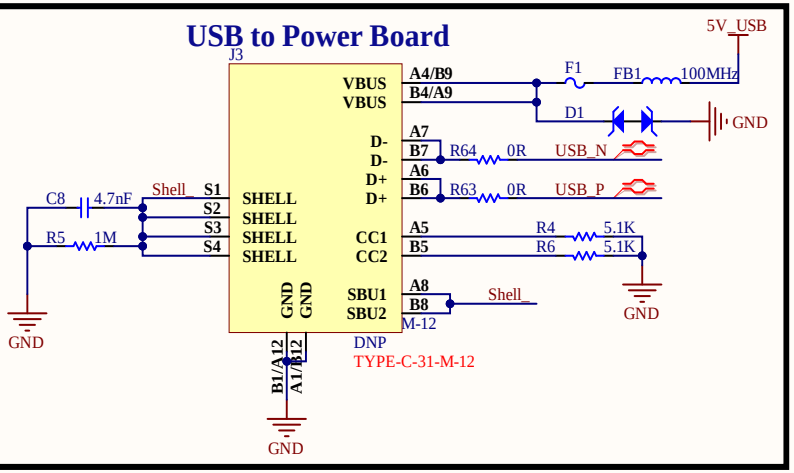
Option 3: (Dual Path - LTE and Ethernet both)
---> DNP U17.

Option 4: (Dual Path - 2 x LTE)
---> DNP J20, U12.

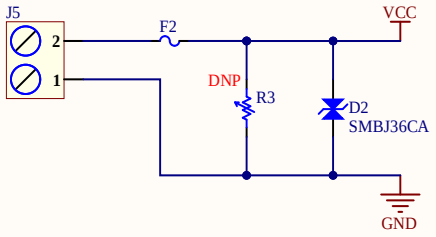
eSpark Consulting Group

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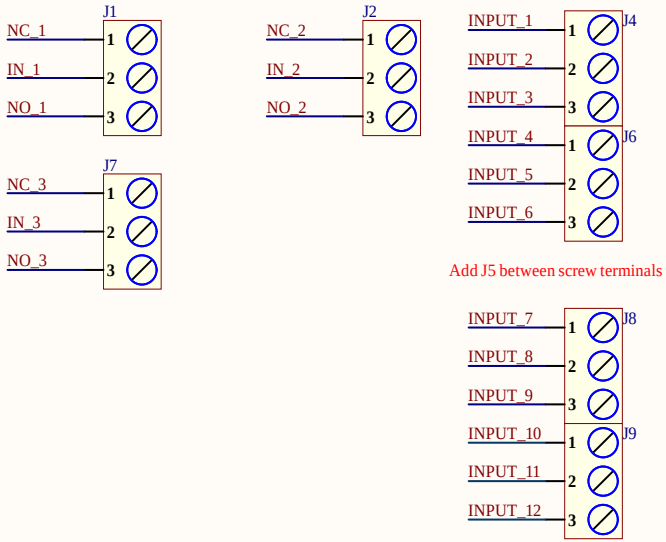
USB to Power Board



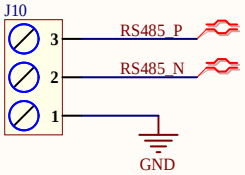
9-30V DC INPUT



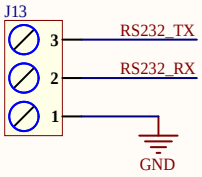
SCREW TERMINALS



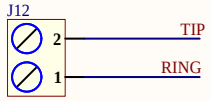
PORT for RS485

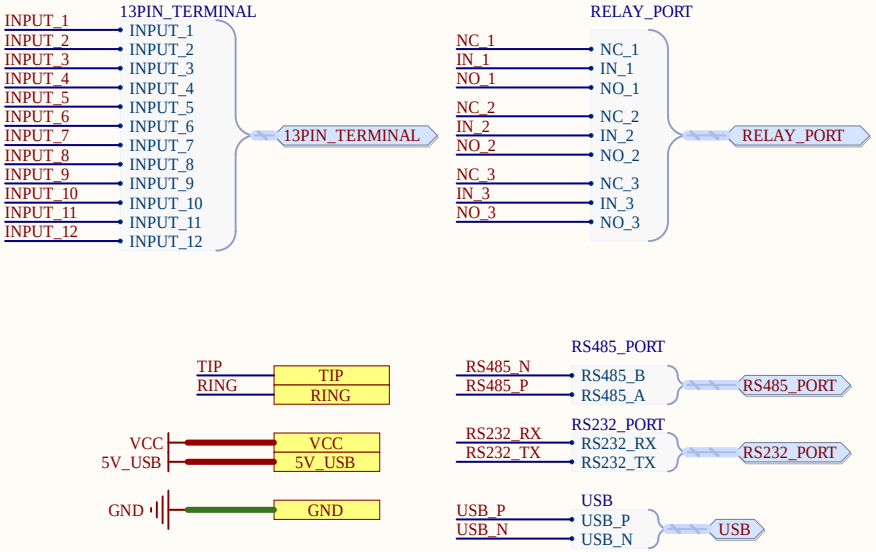


PORT for RS232



PORT for DTMF

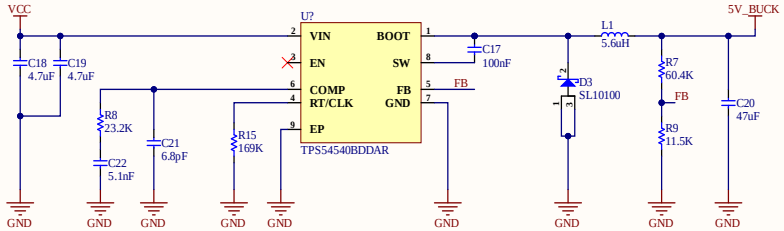




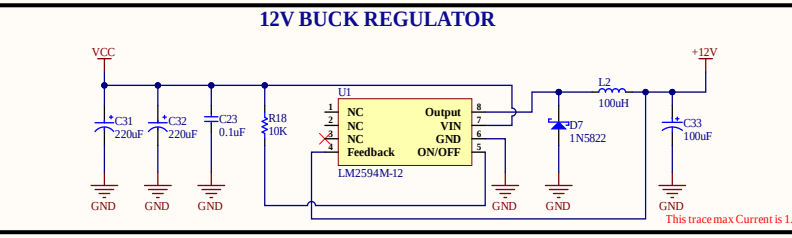
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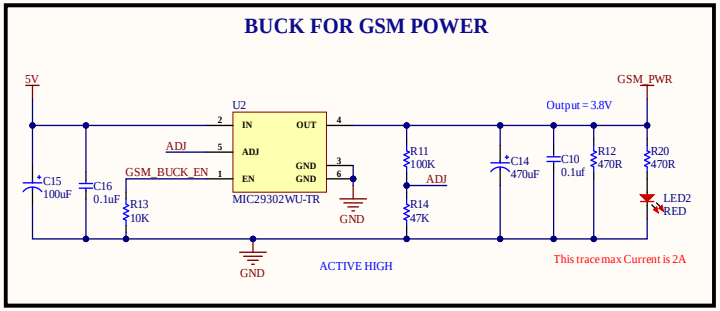
5V BUCK REGULATOR



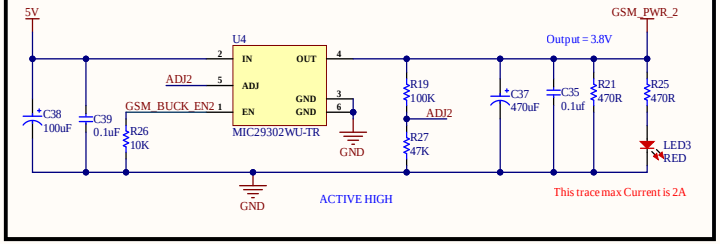
12V BUCK REGULATOR



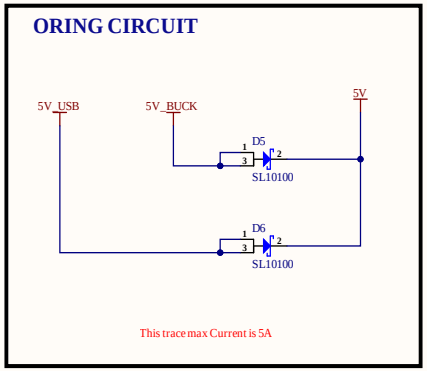
BUCK FOR GSM POWER



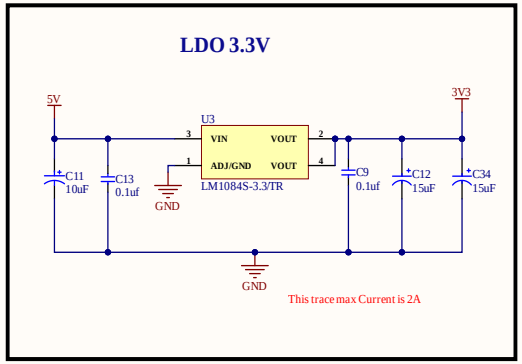
BUCK 2 FOR GSM POWER 2



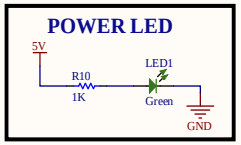
ORING CIRCUIT



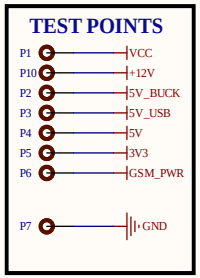
LDO 3.3V

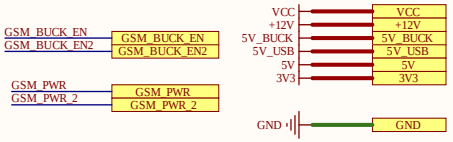


POWER LED

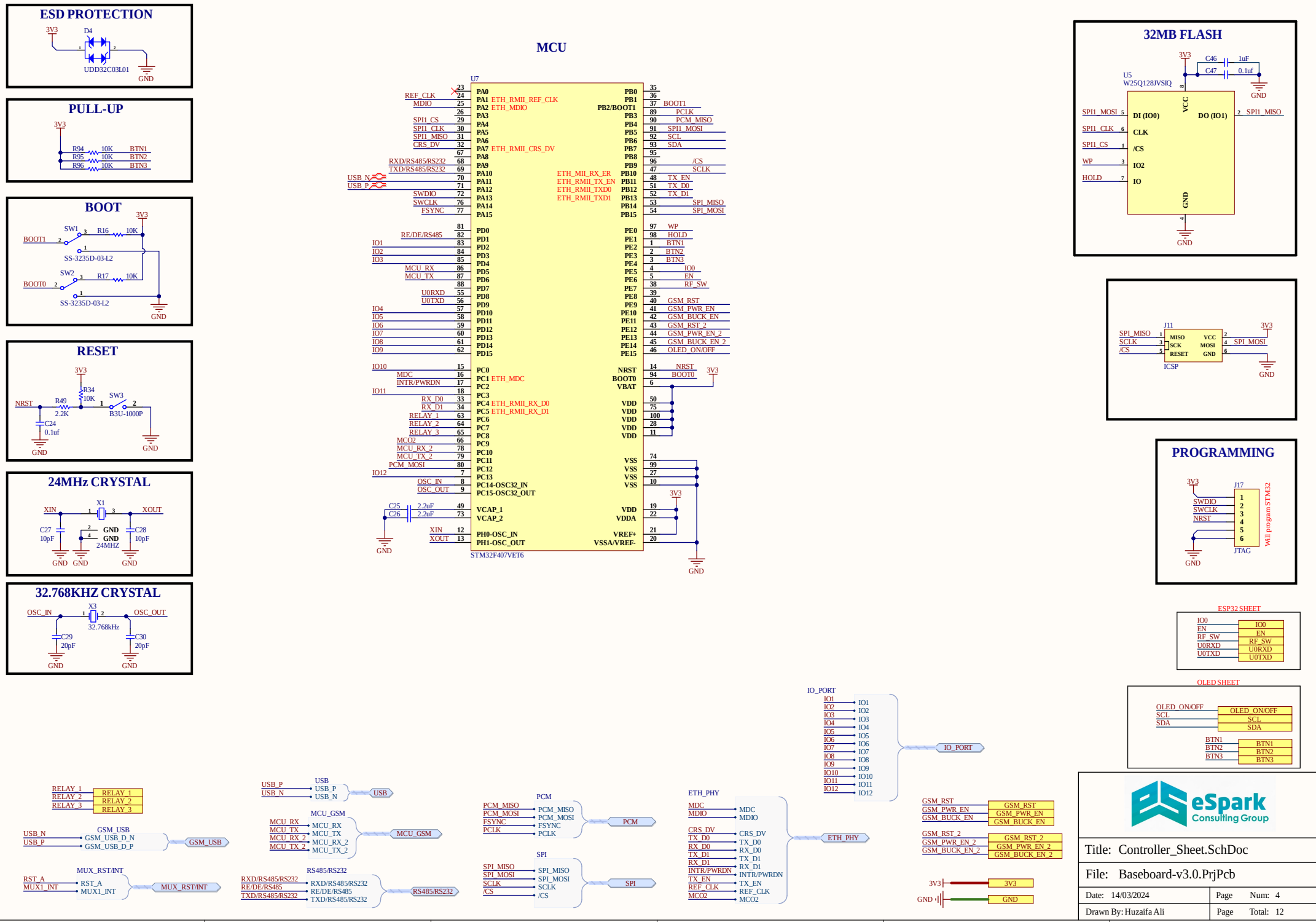


TEST POINTS

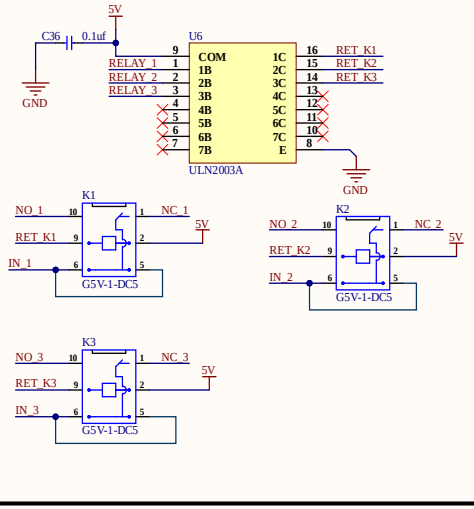




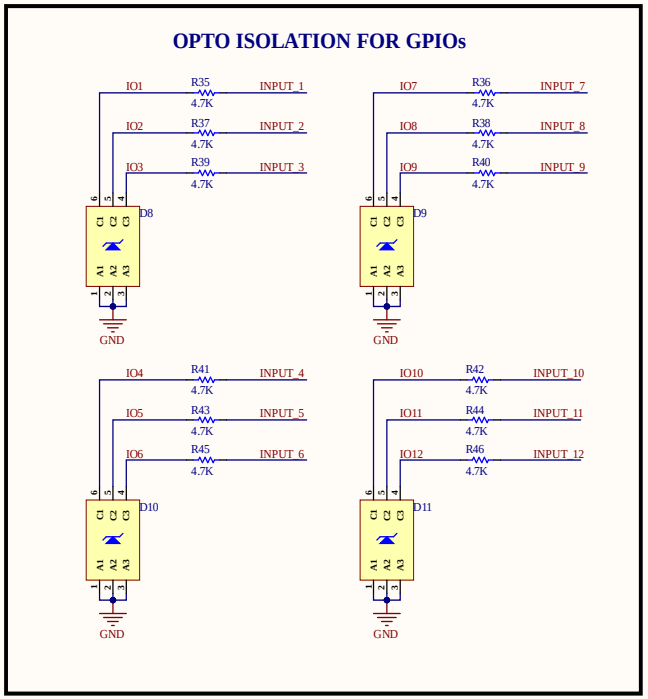
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ESD PROTECTION
3V3
D4
UDD32C03L01
GND
PULL-UP
3V3
R94 10K BTN1
R95 10K BTN2
R96 10K BTN3
BOOT
3V3
SW1
R16 10K
BOOT1
SS-3235D-03-L2
SW2
R17 10K
BOOT0
SS-3235D-03-L2
GND
RESET
3V3
R34 10K
NRST
R49 2.2K
SW3
B3U-1000P
C24 0.1uf
GND
24MHz CRYSTAL
X1
XIN
XOUT
C27 10pF
C28 10pF
GND
24MHZ
32.768KHZ CRYSTAL
X3
OSC_IN
OSC_OUT
32.768kHz
C29 20pF
C30 20pF
MCU
U7
STM32F407VET6
PA0
PA1 ETH_RMII_REF_CLK
PA2 ETH_MDIO
PA3
PA4
PA5
PA6
PA7 ETH_RMII_CRS_DV
PA8
PA9
PA10
PA11
PA12
PA13
PA14
PA15
PB0
PB1
PB2/BOOT1
PB3
PB4
PB5
PB6
PB7
PB8
PB9
PB10
PB11
PB12
PB13
PB14
PB15
PC0
PC1 ETH_MDC
PC2
PC3
PC4 ETH_RMII_RX_D0
PC5 ETH_RMII_RX_D1
PC6
PC7
PC8
PC9
PC10
PC11
PC12
PC13
PC14-OSC32_IN
PC15-OSC32_OUT
PD0
PD1
PD2
PD3
PD4
PD5
PD6
PD7
PD8
PD9
PD10
PD11
PD12
PD13
PD14
PD15
PE0
PE1
PE2
PE3
PE4
PE5
PE6
PE7
PE8
PE9
PE10
PE11
PE12
PE13
PE14
PE15
NRST
BOOT0
VBAT
VDD
VSS
VCAP_1
VCAP_2
VDDA
VREF+
VSSA/VREF-
PH0-OSC_IN
PH1-OSC_OUT
REF_CLK
MDIO
SPI1_CS
SPI1_CLK
SPI1_MISO
CRS_DV
RXD/RS485/RS232
TXD/RS485/RS232
USB_N
USB_P
SWDIO
SWCLK
FSYNC
RE/DE/RS485
IO1
IO2
IO3
MCU_RX
MCU_TX
U0RXD
U0TXD
IO4
IO5
IO6
IO7
IO8
IO9
IO10
MDC
INTR/PWRDN
IO11
RX_D0
RX_D1
RELAY_1
RELAY_2
RELAY_3
MCO2
MCU_RX_2
MCU_TX_2
PCM_MOSI
IO12
OSC_IN
OSC_OUT
XIN
XOUT
BOOT1
PCLK
PCM_MISO
SPI1_MOSI
SCL
SDA
/CS
SCLK
TX_EN
TX_D0
TX_D1
SPI_MISO
SPI_MOSI
WP
HOLD
IO0
EN
RF_SW
GSM_RST
GSM_PWR_EN
GSM_BUCK_EN
GSM_RST_2
GSM_PWR_EN_2
GSM_BUCK_EN_2
OLED_ON/OFF
BTN1
BTN2
BTN3
ETH_MII_RX_ER
ETH_RMII_TX_EN
ETH_RMII_TXD0
ETH_RMII_TXD1
32MB FLASH
U5
W25Q128JVSIQ
DI (IO0)
DO (IO1)
CLK
/CS
IO2
IO
VCC
GND
C46 1uF
C47 0.1uf
ICSP
J11
MISO
SCK
RESET
MOSI
PROGRAMMING
J17
JTAG
Will program STM32
ESP32 SHEET
OLED SHEET
IO_PORT
ETH_PHY
PCM
SPI
USB
MCU_GSM
GSM_USB
MUX_RST/INT
RS485/RS232
RELAY_1
RELAY_2
RELAY_3
GSM_USB_D_N
GSM_USB_D_P
RST_A
MUX1_INT
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Page Total: 12
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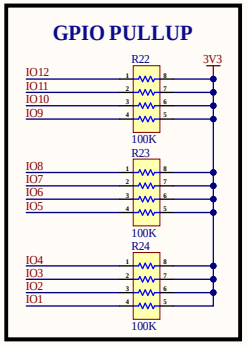
RELAY WITH DRIVER

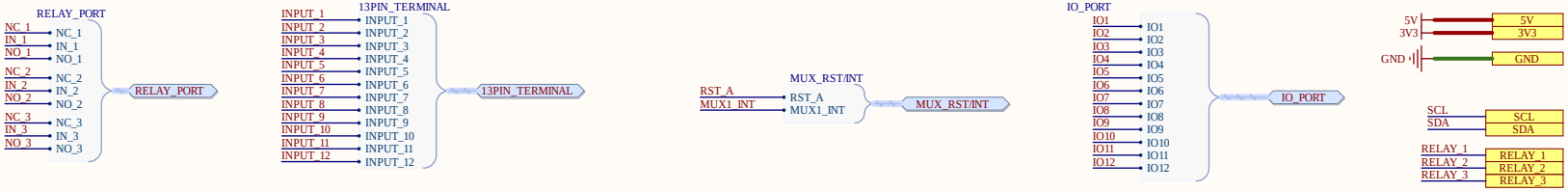


OPTO ISOLATION FOR GPIOs



GPIO PULLUP

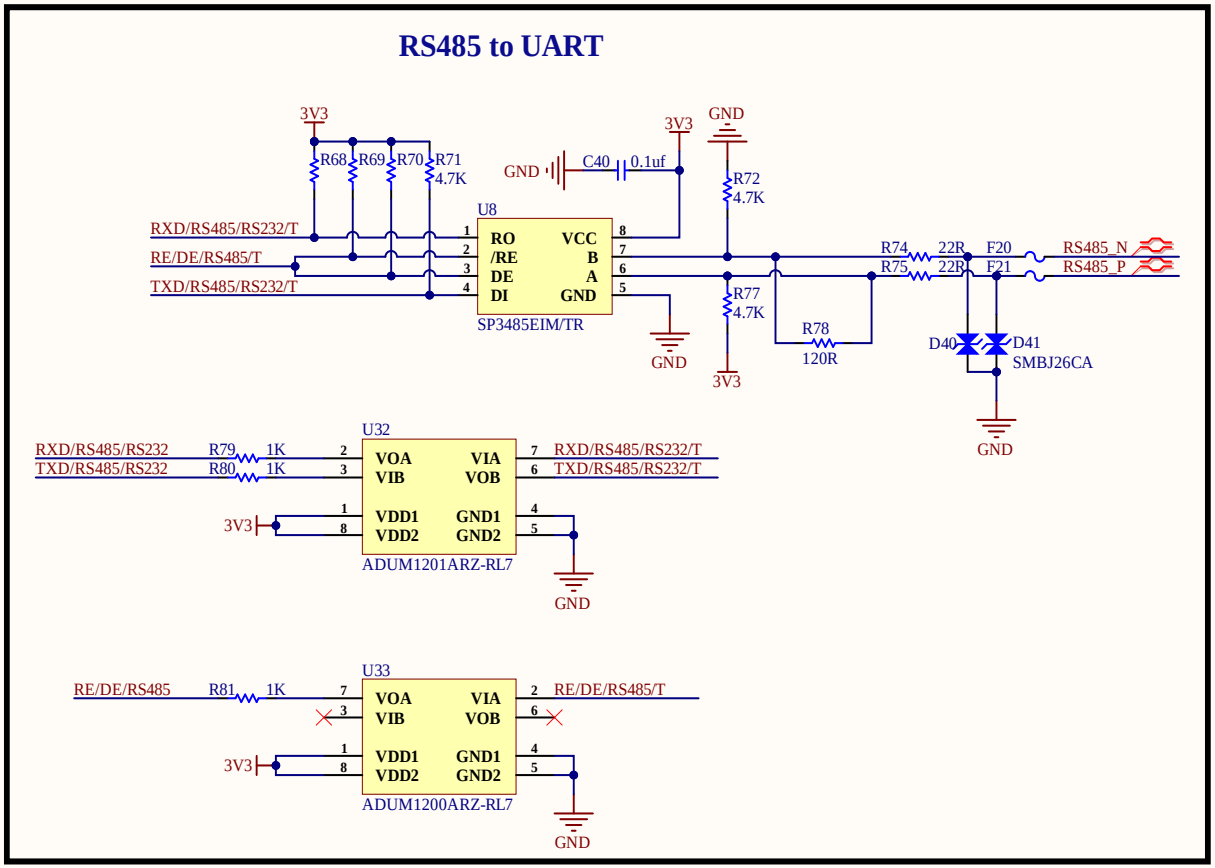




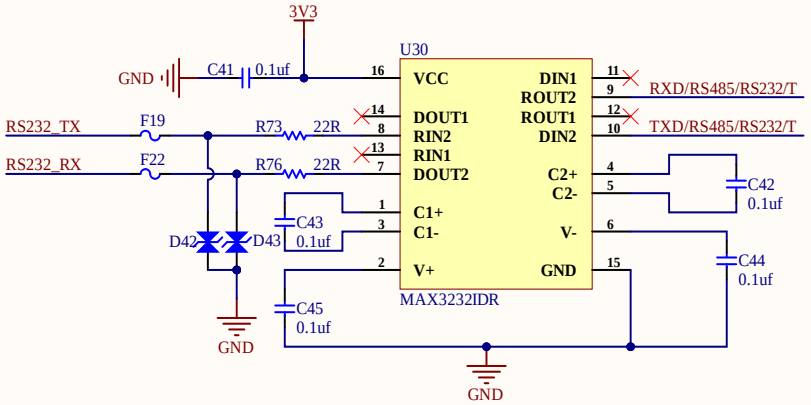
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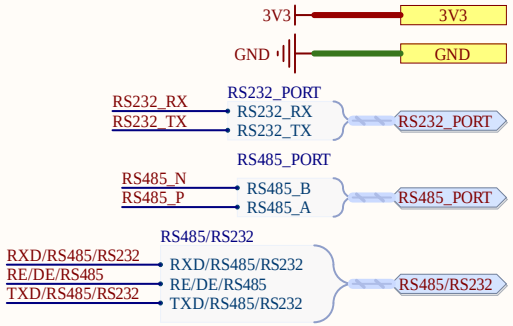
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RS485 to UART



RS232 to UART

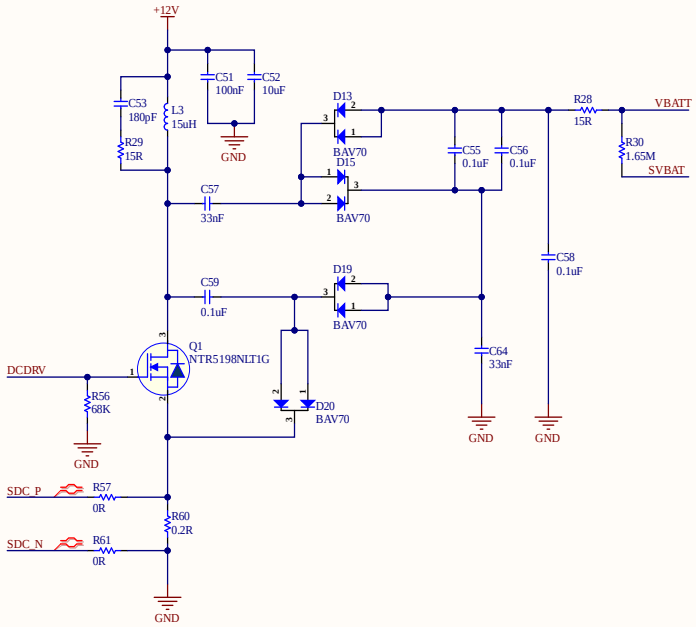




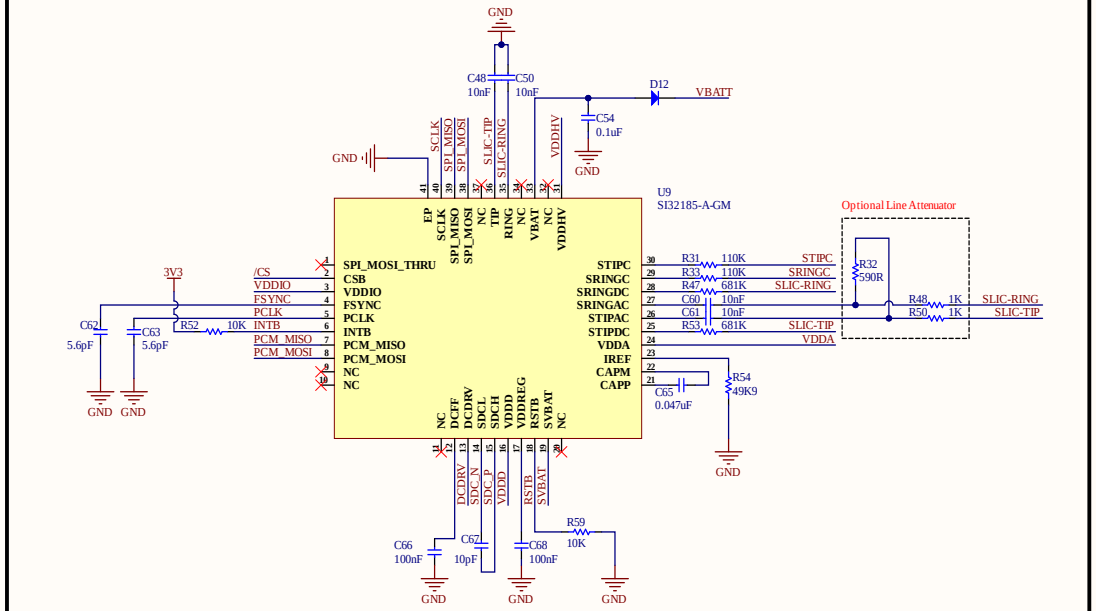
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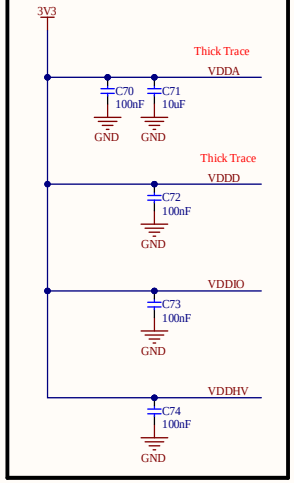
CAPACITIVE BOOST

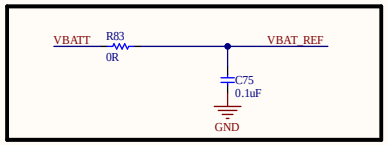


SI32185

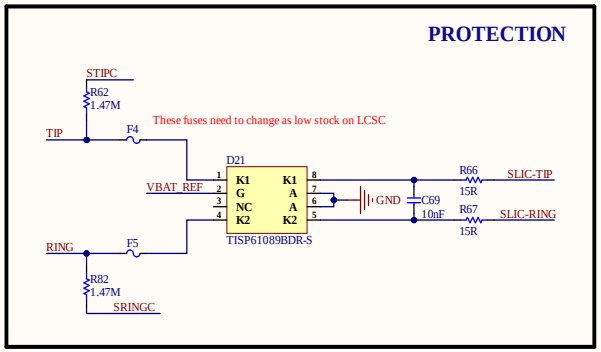


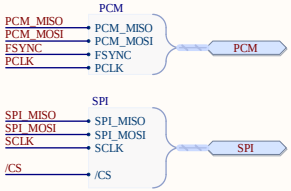
POWER

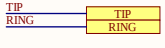


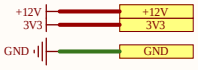


PROTECTION





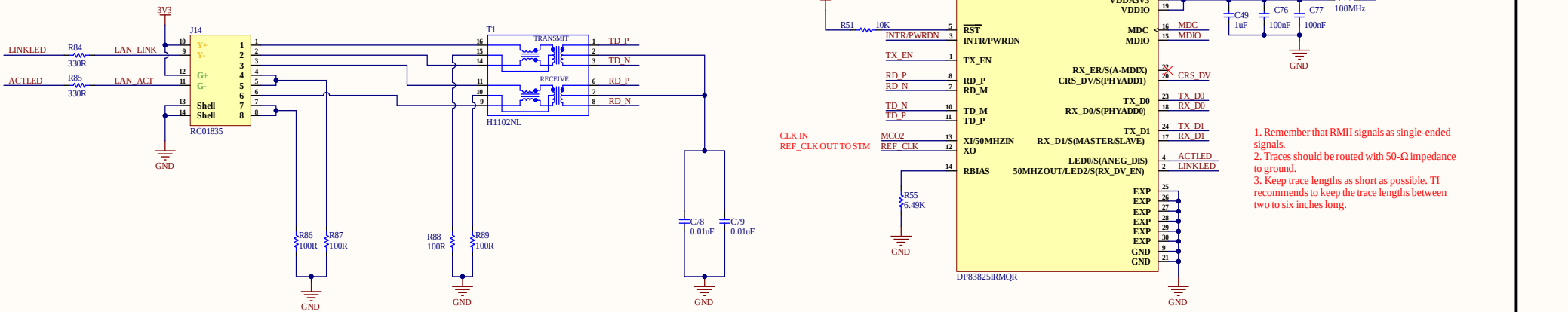




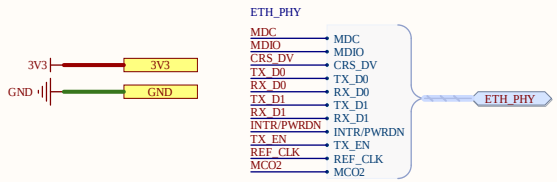
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ETHERNET/ DP83825IRMQR PHY

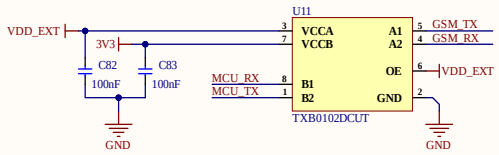


1. Remember that RMII signals as single-ended signals.
2. Traces should be routed with 50-Ω impedance to ground.
3. Keep trace lengths as short as possible. TI recommends to keep the trace lengths between two to six inches long.

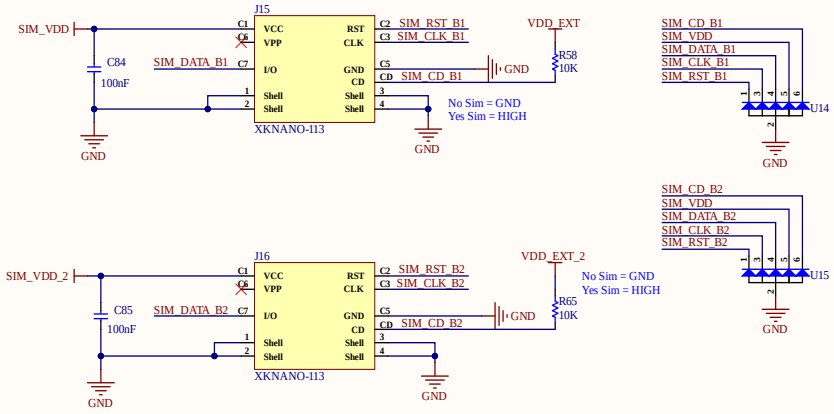


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Level Shifter for GSM Serial 1

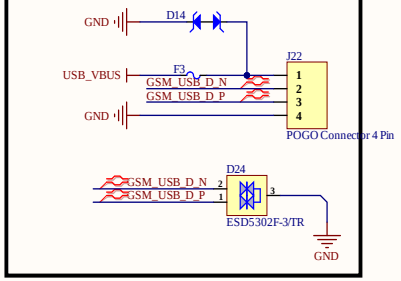


DUALSIM WITH MULTIPLEXER

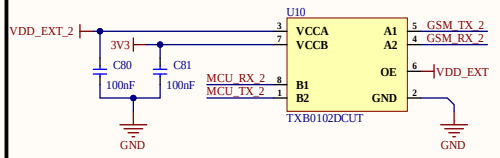


SIM Card Design Guide
SIM card signal could be interferenced by some high frequency signal, it is strongly recommendedtofollowthese guidelines while designing:
- SIM card holder should be far away from main antenna
- SIM traces should keep away from RF lines, VBAT and high-speed signal lines
- The traces should be as short as possible
- Keep SIM card holder's GND connect to main ground directly
- Shielding the SIM card signal by ground well
- Recommended to place a 100nF capacitor on SIM_VDD line and keep close to the SIMcardholder
- Add some TVS which parasitic capacitance should not exceed 50pF
- Add 22Ω resistor to (SIM_RST/SIM_CLK/SIM_DATA) signal could enhance ESDprotection
- Mount 22pF capacitor to (SIM_RST/SIM_CLK/SIM_DATA) signal if having RF signal interference

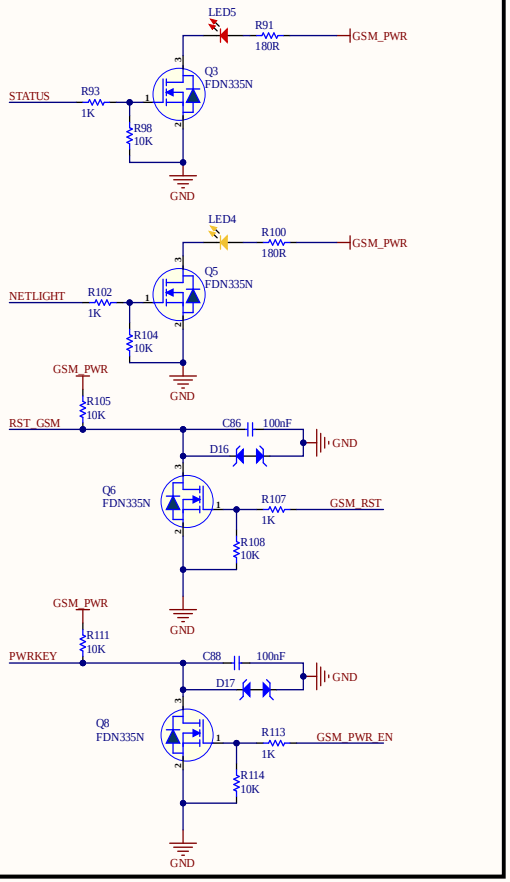
USB POGO for Updates



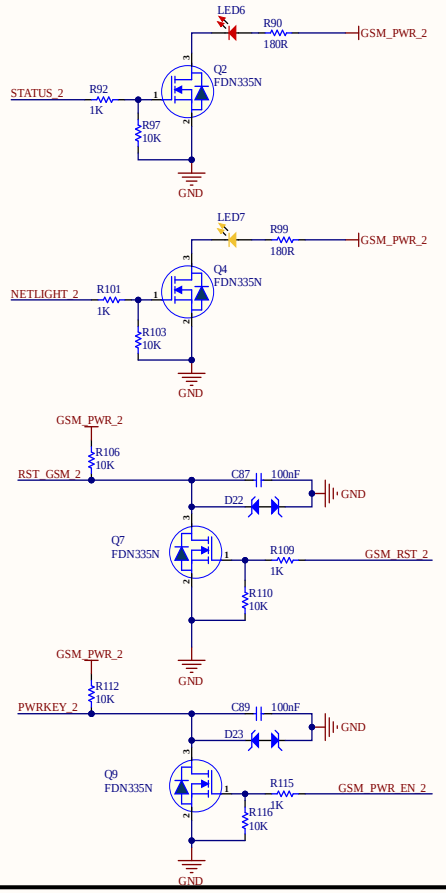
Level Shifter for GSM Serial 2

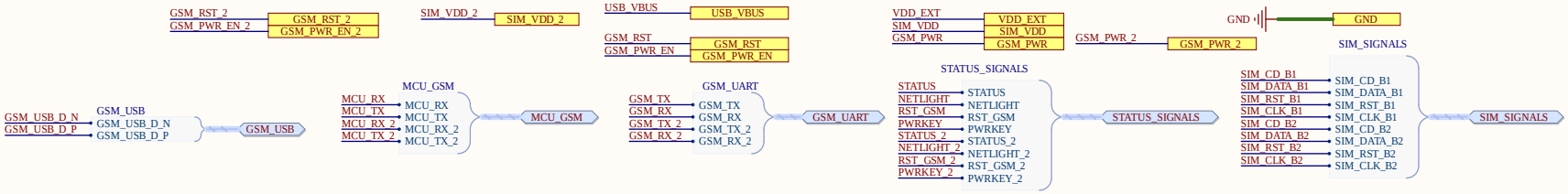


GSM_COMMANDS_1



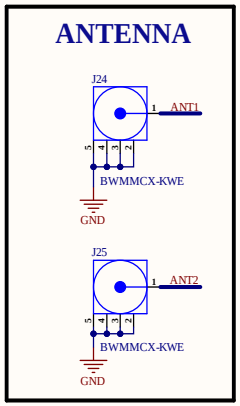
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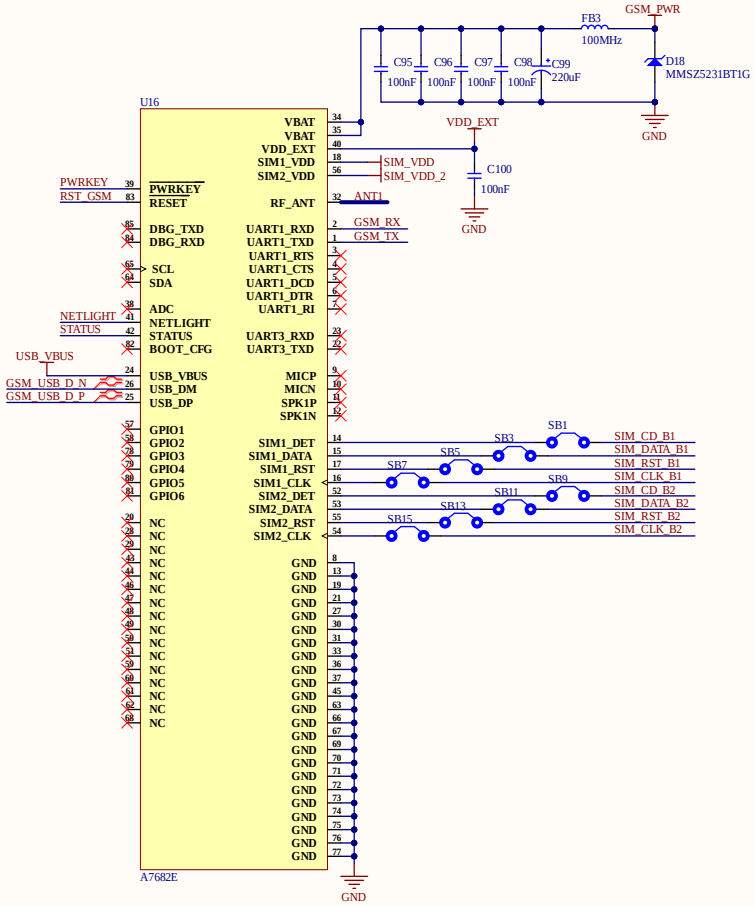


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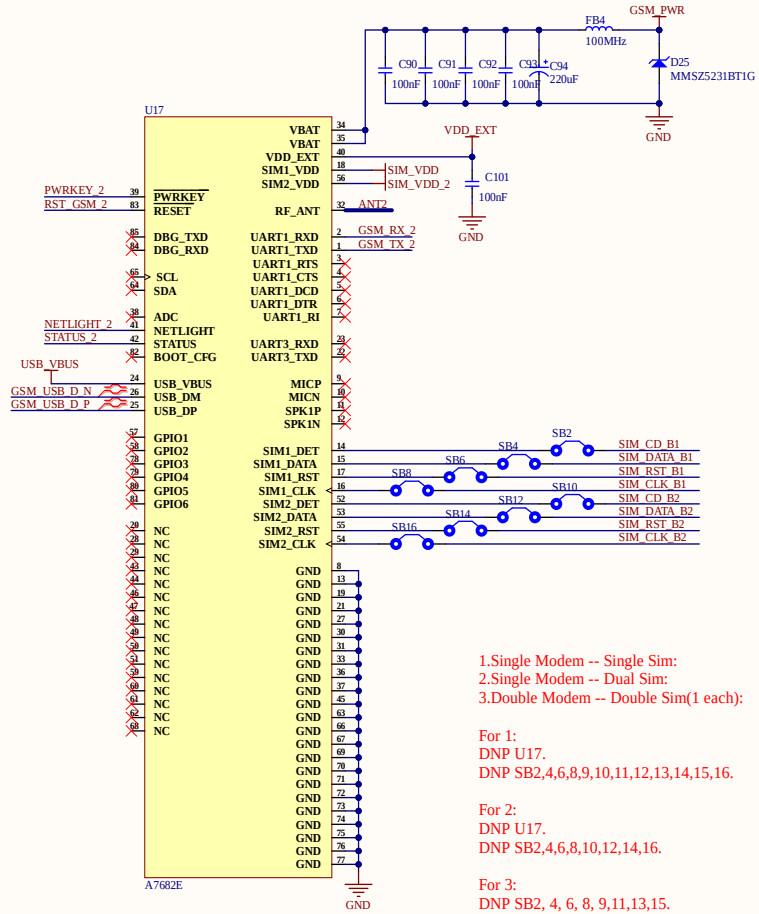
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Date: 14/03/2024	Page Num: 9
Drawn By: Huzaifa Ali	Page Total: 12



A7682E 01



A7682E 02

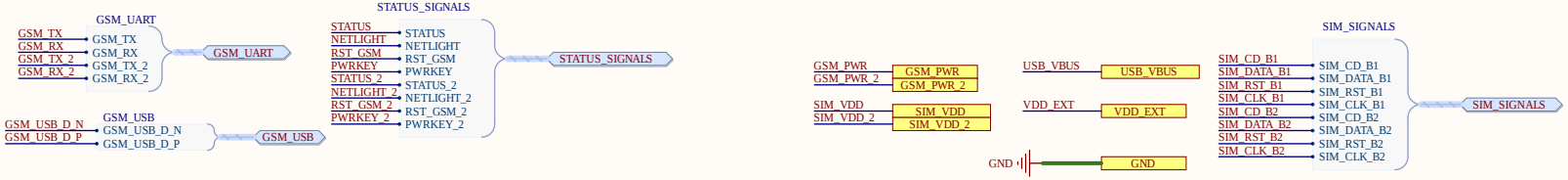


1.Single Modem -- Single Sim:
2.Single Modem -- Dual Sim:
3.Double Modem -- Double Sim(1 each):

For 1:
DNP U17.
DNP SB2,4,6,8,9,10,11,12,13,14,15,16.

For 2:
DNP U17.
DNP SB2,4,6,8,10,12,14,16.

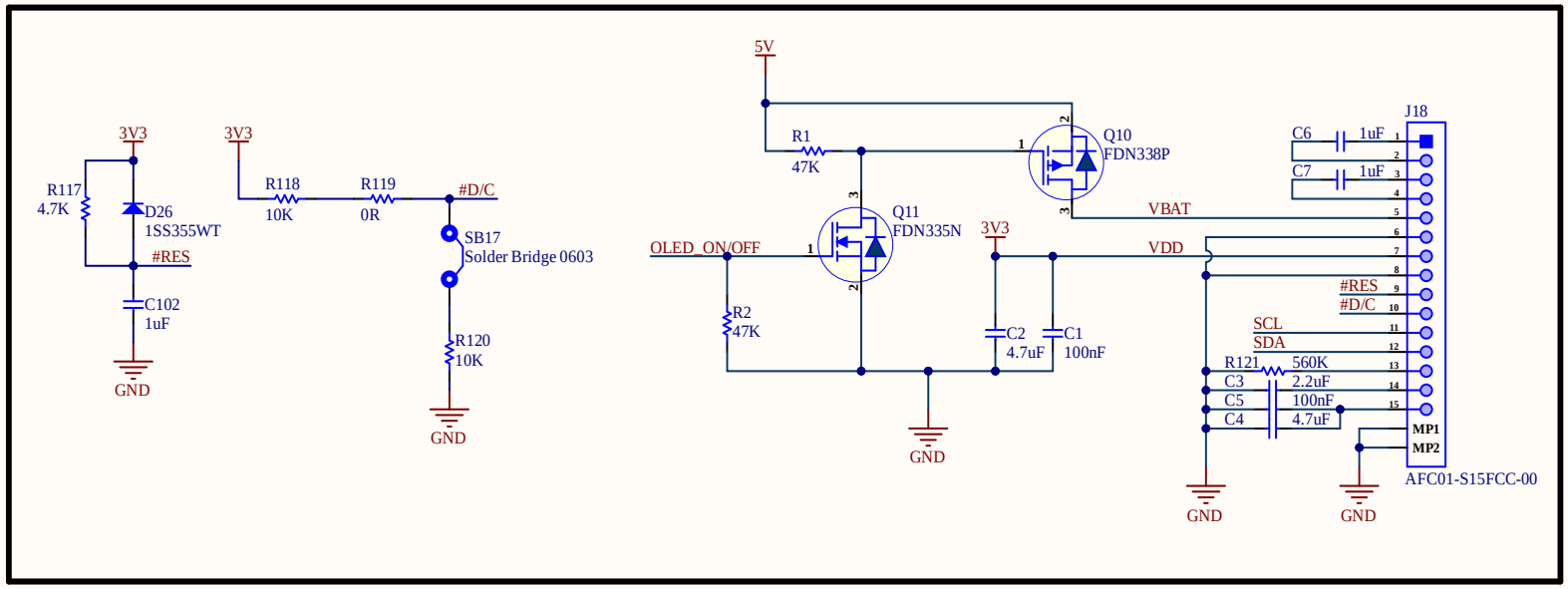
For 3:
DNP SB2, 4, 6, 8, 9,11,13,15.

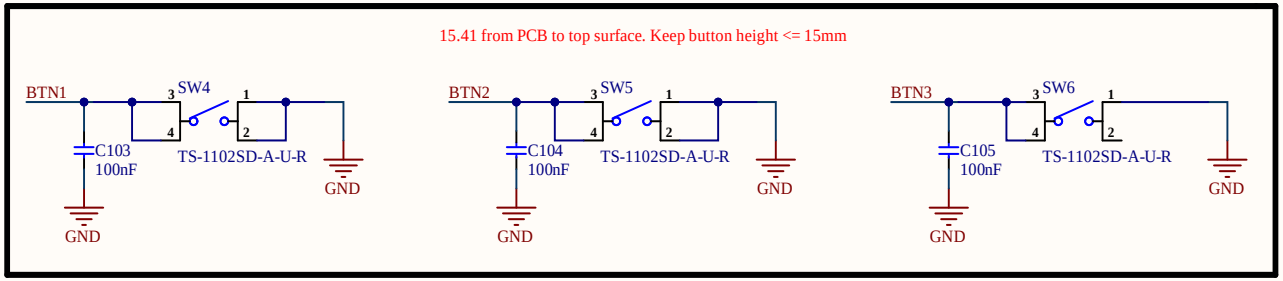


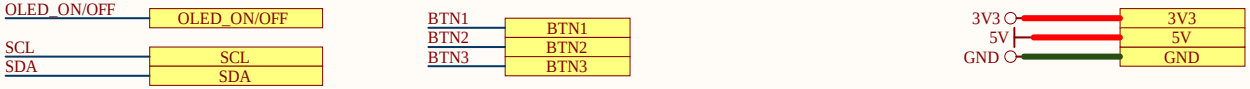
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OLED

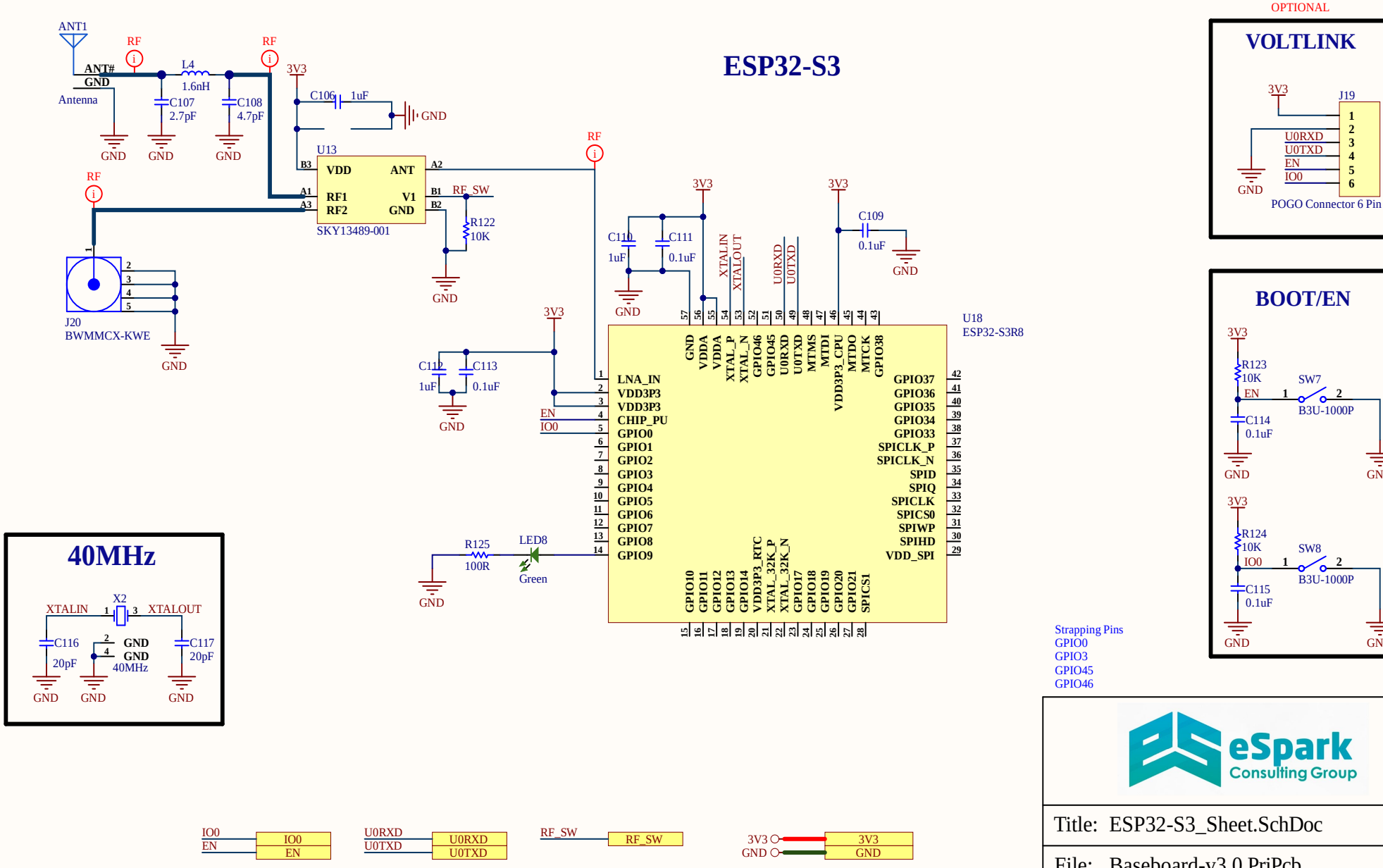






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